

Bangladesh Army University of Science and Technology (BAUST), Saidpur



Project Report

Department : Computer Science and Engineering (CSE)

Course Title : Database Management Systems Sessional

Course No : CSE 2206

Name of Topic : High School Management System

Level : 2 **Term** : II

Submitted By:

ID	Name
210201020	Ifti Haque
210201023	Nihad Hossain
210201044	Md. Nur A Alam
210201045	Rakib Hasan Fahim

Submitted To:

Name of Teacher : **Hasan Muhammad Kafi**

Designation : **Assistant Professor**

Entity list:

1. Student
2. Guardian
3. Teacher
4. Result
5. Staff
6. Department
7. Subject
8. Class
9. Exam
10. Attendance
11. Schedule
12. Accounting

Entity set attributes:**1. Student**St_ID

St_Name

Gender

Blood_group

Date_of_Birth

Age()

Address

Contact_Details

{phone_number}

Email address

Guardian_ID

Admission_Details

Date_of_admission

Class

Section

St_Picture

2. GuardianGuardian_ID

Guardian_name

St_ID

National_ID

Address

Contact_Details

Phone_number

Email_address

Student_Relationship

3. TeacherTeacher_ID

Teacher_Name

Gender

Blood_group

Date_of_Birth

Age()

Address

Contact_Details

Phone_number

Email_address

National_ID

Education_Qualification

Degree

Department

Result

Employment_Details

Joining_date

Position

Salary

Teacher_Picture

4. ResultResult_ID

Exam_ID

Subject_ID

Marks_obtain

Grade

GPA

5. Staff

Staff_ID
Staff_Name
Gender
Blood_group
Date_of_Birth
Age()
Address
Contact_Details
 Phone_number
 Email_address
National_ID
Education_Qualification
 Degree
 Department
 Result
Employment_Details
 Joining_date
 Position
 Salary
Staff_Picture

6. Department

Department_ID
Department_Name
Faculty_list
Employment_Details
Teacher_ID
Staff_ID

7. Subject

Subject_ID
Subject_name
Class_ID
Teacher_ID

8. Class

Class_ID
Class_name
Assign_Teacher
 Teacher_ID
 Subject_ID
Student_ID

9. Exam

Exam_ID
Class_ID
Subject_ID
Date
Time

10. Attendance

Date
Student_ID
Teacher_ID
Class_ID

11. Schedule

Class_ID
Subject_ID
Teacher_ID
Time
Week_days
Holidays

12. Accounting

Accountant_ID
Student_ID
Guardian_ID
Tution_Fee
 Date
 Amount
Payment_Status

Relationship Set

❖ Student - Guardian (One-to-Many)

- Name: **Parent**
- One student can have one or more parents/guardians.
- The Parent/Guardian_ID in the Student entity is a foreign key that references the primary key Parent_ID in the Parent / Guardian entity.

❖ Teacher - Department (Many-to-Many)

- Name: **Employe**
- A teacher can work in many departments.
- A department can have many teachers.
- A join table is needed to represent this relationship.

❖ Teacher - Class (Many-to-Many)

- Name: **Tech Class**
- A teacher can teach many classes.
- A class can have many teacher.
- The TeacherID in the Class entity is a foreign key that references the primary key TeacherID in the Teacher entity.

❖ Student - Class (One-to-Many)

- Name: **Stu Class**
- Many students can be enrolled in one class.
- A class can have many students.
- The ClassID in the Student entity is a foreign key that references the primary key ClassID in the Class entity.

❖ Staff - Department (Many-to-Many)

- Name: **Employe**
- A staff member can work in many departments.
- A department can have many staff members.
- A join table is needed to represent this relationship.

❖ Department - Subject (One-to-Many)

- Name: **Dept Sub**
- A department can offer many subjects.
- A subject can belong to only one department.

- The DepartmentID in the Subject entity is a foreign key that references the primary key DepartmentID in the Department entity.

❖ **Exam - Class (Many-to-Many)**

- Name: **Exm Class**
- An exam can be taken by many classes.
- A class can take many exams.
- The ClassID in the Exam entity is a foreign key that references the primary key ClassID in the Class entity.

❖ **Exam - Subject (Many-to-Many)**

- Name: **Exm Sub**
- An exam can be taken for many subjects.
- A subject can have many exams.
- The SubjectID in the Exam entity is a foreign key that references the primary key SubjectID in the Subject entity.

❖ **Result - Exam (One-to-One)**

- Name: **Res Exm**
- One result corresponds to one exam only.
- The ExamID in the Result entity is a foreign key that references the primary key ExamID in the Exam entity.

❖ **Result - Student (One-to-One)**

- Name: **Grade**
- One result corresponds to one student only.
- The St_ID in the Result entity is a foreign key that references the primary key St_ID in the Student entity.

❖ **Attendance – Student (Many-to-One)**

- Name: **Atnd Stu**
- Many students can attend one class
- One student attends only one class
- The Class_ID in Attendance entity is a foreign key that references the primary key Class_ID in Class entity
- The St_ID in Attendance entity is a foreign key that references the primary key St_ID in Student entity.

❖ **Attendance – Teacher (Many-to-One)**

- Name: **Atnd Tech**

- Many teachers can take attendance for one class
- One teacher takes attendance for only one class
- The Class_ID in Attendance entity is a foreign key that references the primary key Class_ID in Class entity
- The Teacher_ID in Attendance entity is a foreign key that references the primary key Teacher_ID in Teacher entity

❖ **Schedule – Student (One-to-Many)**

- Name: **Routine**
- One student can do only one class at a time
- One class can be done by many students at a time
- The Student_ID and Subject_ID both are foreign keys that reference their respective primary keys of Teacher and Subject entities respectively.

❖ **Schedule – Teacher (One-to-One)**

- Name: **Routine**
- One teacher teaches only one subject at a time
- One subject is taught by only one teacher at a time
- The Teacher_ID and Subject_ID both are foreign keys that reference their respective primary keys of Teacher and Subject entities respectively.

❖ **Accounting - Guardian (One-to-Many)**

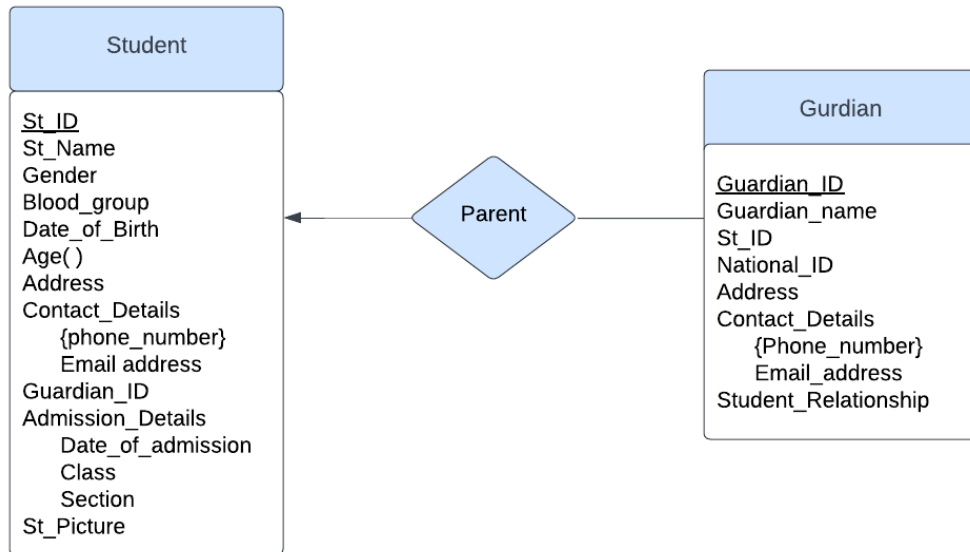
- Name: **payment**
- One guardian can pay for one or more students.
- A student can have only one guardian.
- The GuardianID in the Accounting entity is a foreign key that references the primary key GuardianID in the Guardian entity.

❖ **Accounting - Student (One-to-Many)**

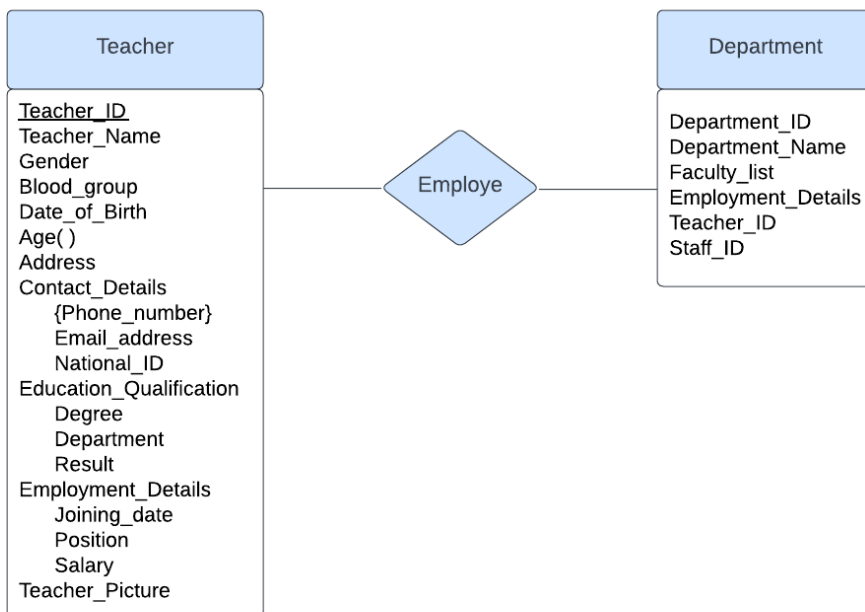
- Name: **payment**
- One student can have one or more accounting records.
- An accounting record corresponds to only one student.
- The St_ID in the Accounting entity is a foreign key that references the primary key St_ID in the Student entity.

Cardinality Mapping:

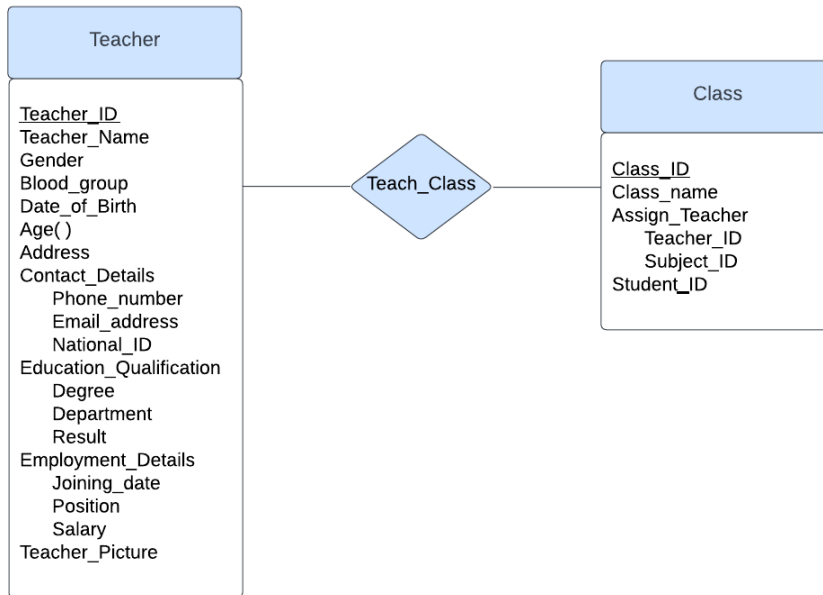
❖ Student - Guardian (One-to-Many)



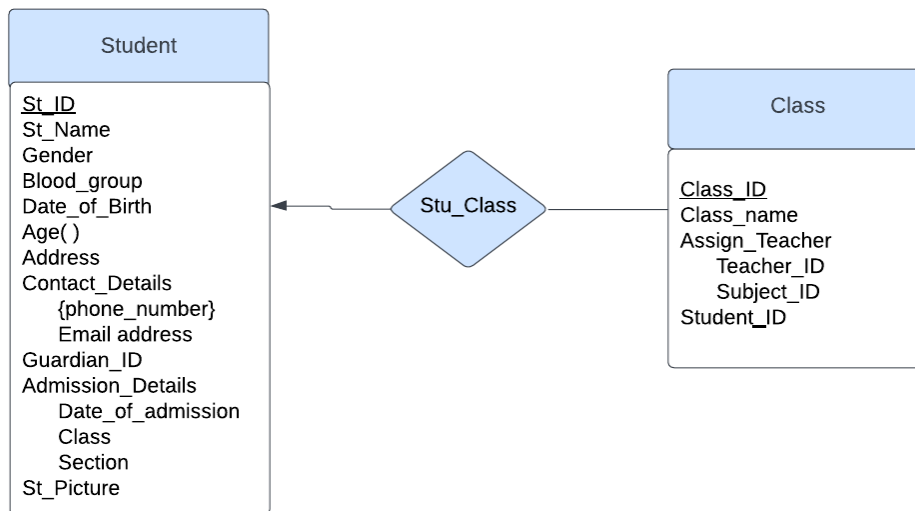
❖ Teacher - Department (Many-to-Many)



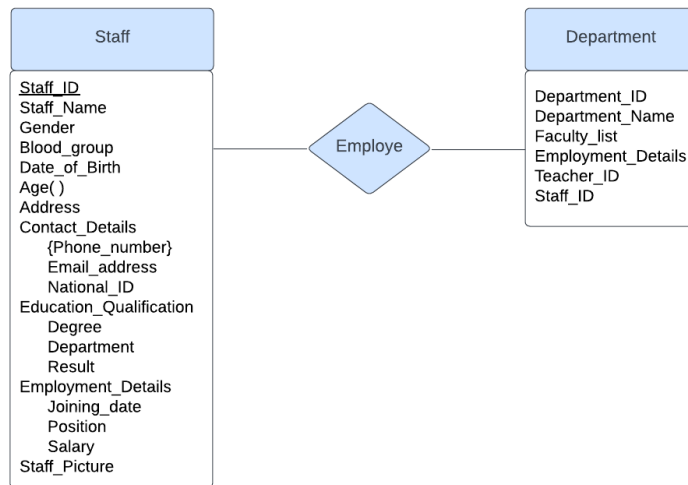
❖ Teacher - Class (Many-to-Many)



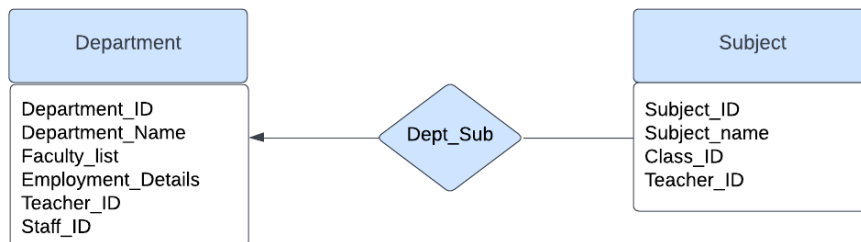
❖ Student - Class (One-to-Many)



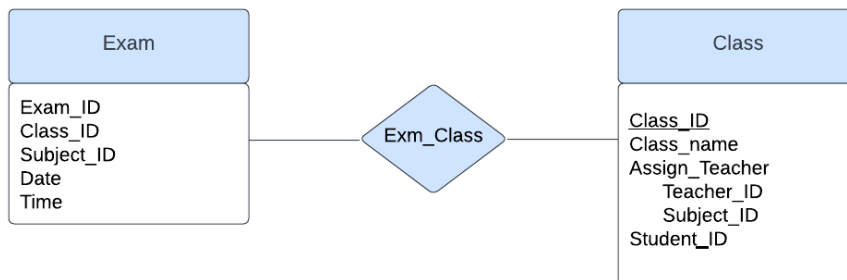
❖ Staff - Department (Many-to-Many)



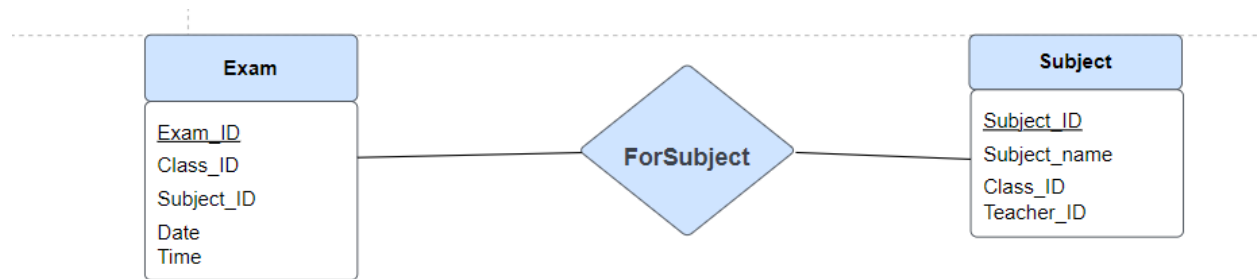
❖ Department - Subject (One-to-Many)



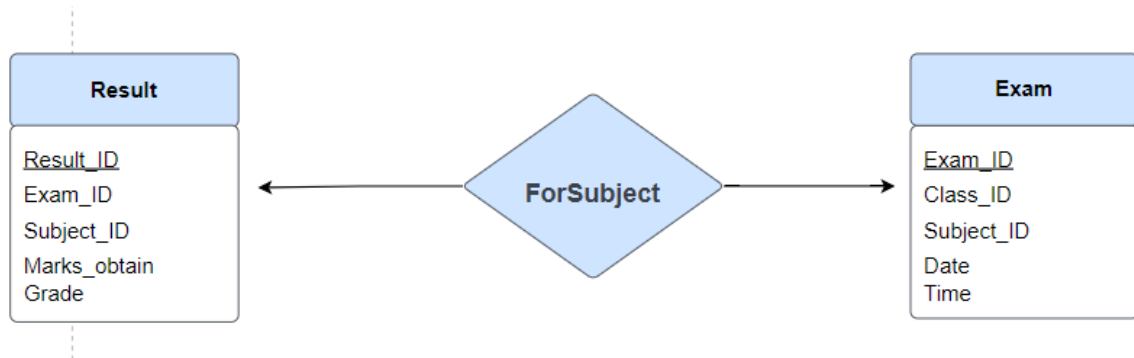
❖ Exam - Class (Many-to-Many)



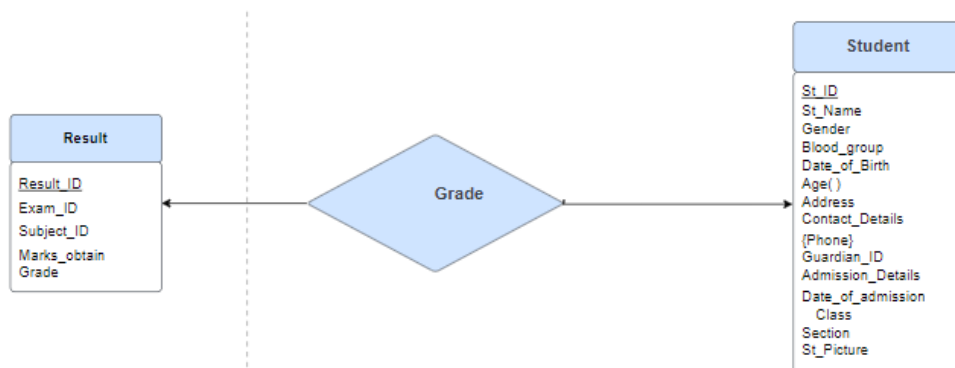
❖ **Exam - Subject (Many-to-Many)**



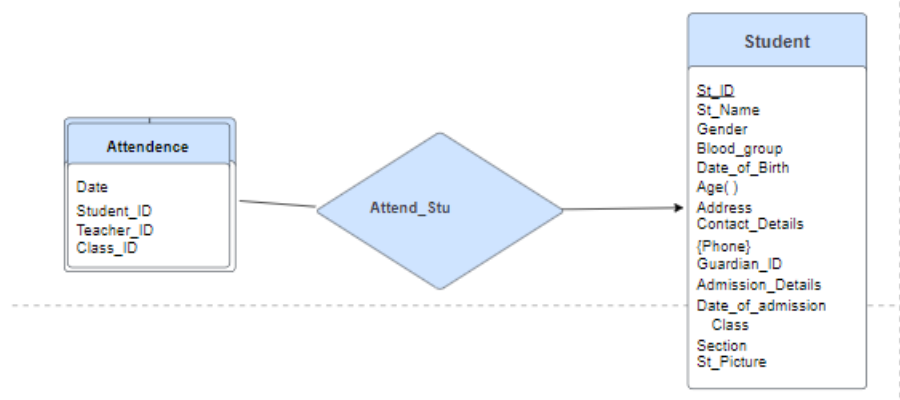
❖ **Result - Exam (One-to-One)**



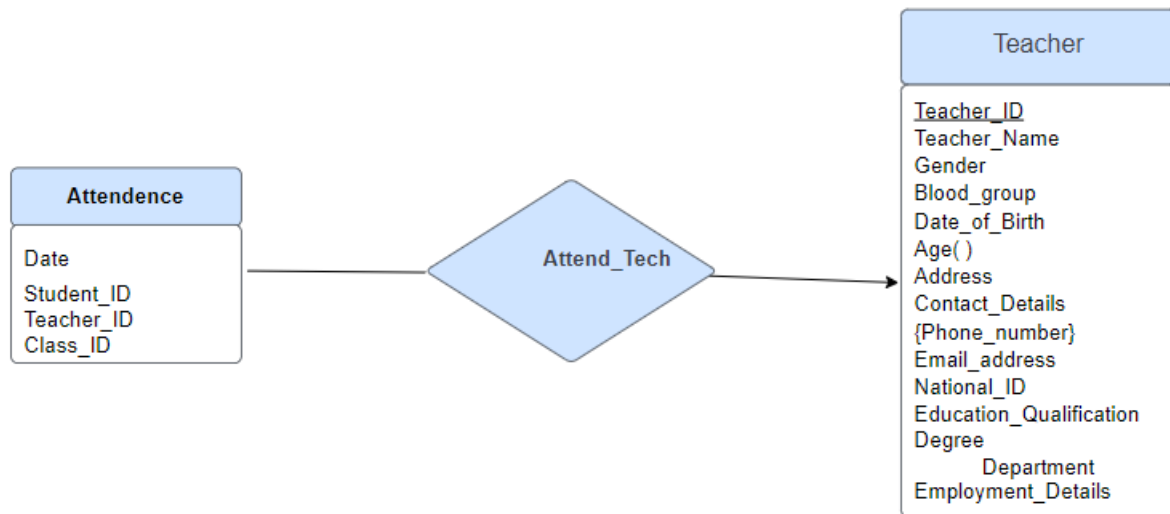
❖ **Result - Student (One-to-One)**



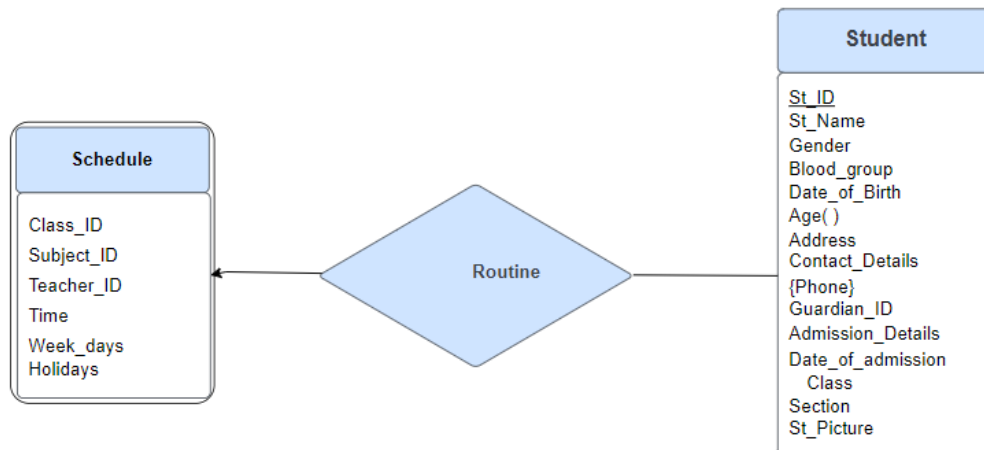
❖ **Attendance – Student (Many-to-One)**



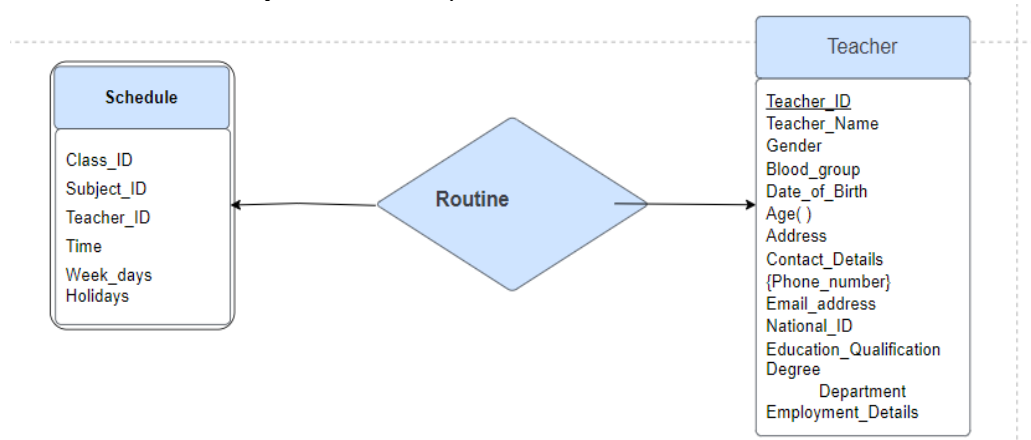
❖ Attendance – Teacher (Many-to-One)



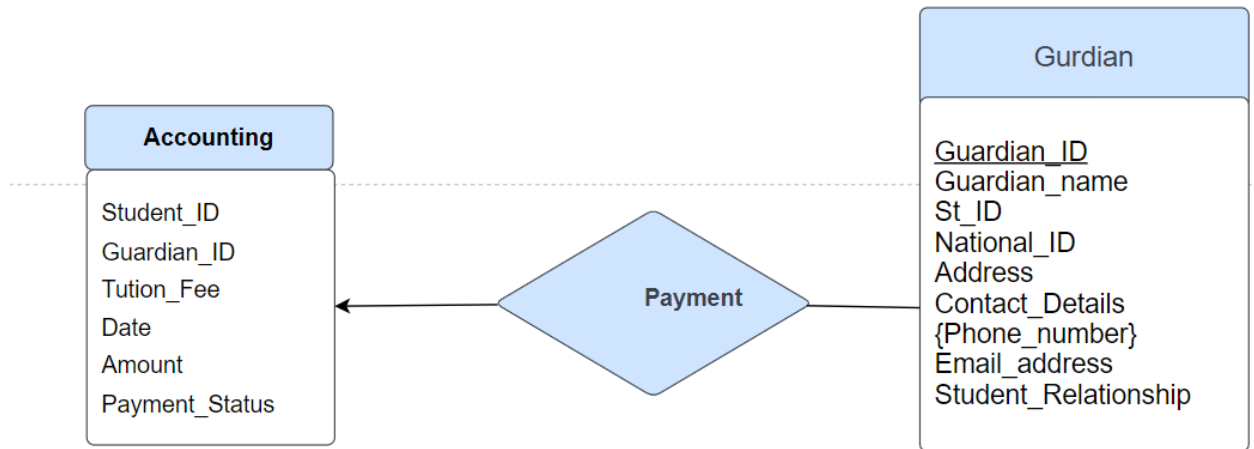
❖ Schedule – Student (One-to-Many)



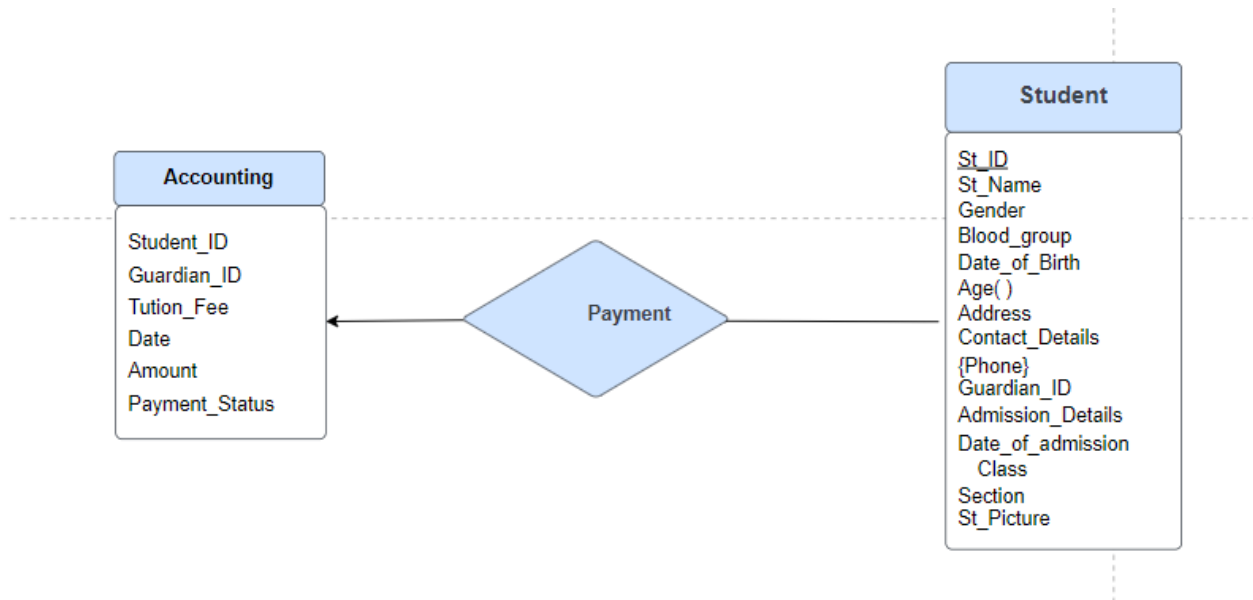
❖ Schedule – Teacher (One-to-One)



❖ **Accounting - Guardian (One-to-Many)**



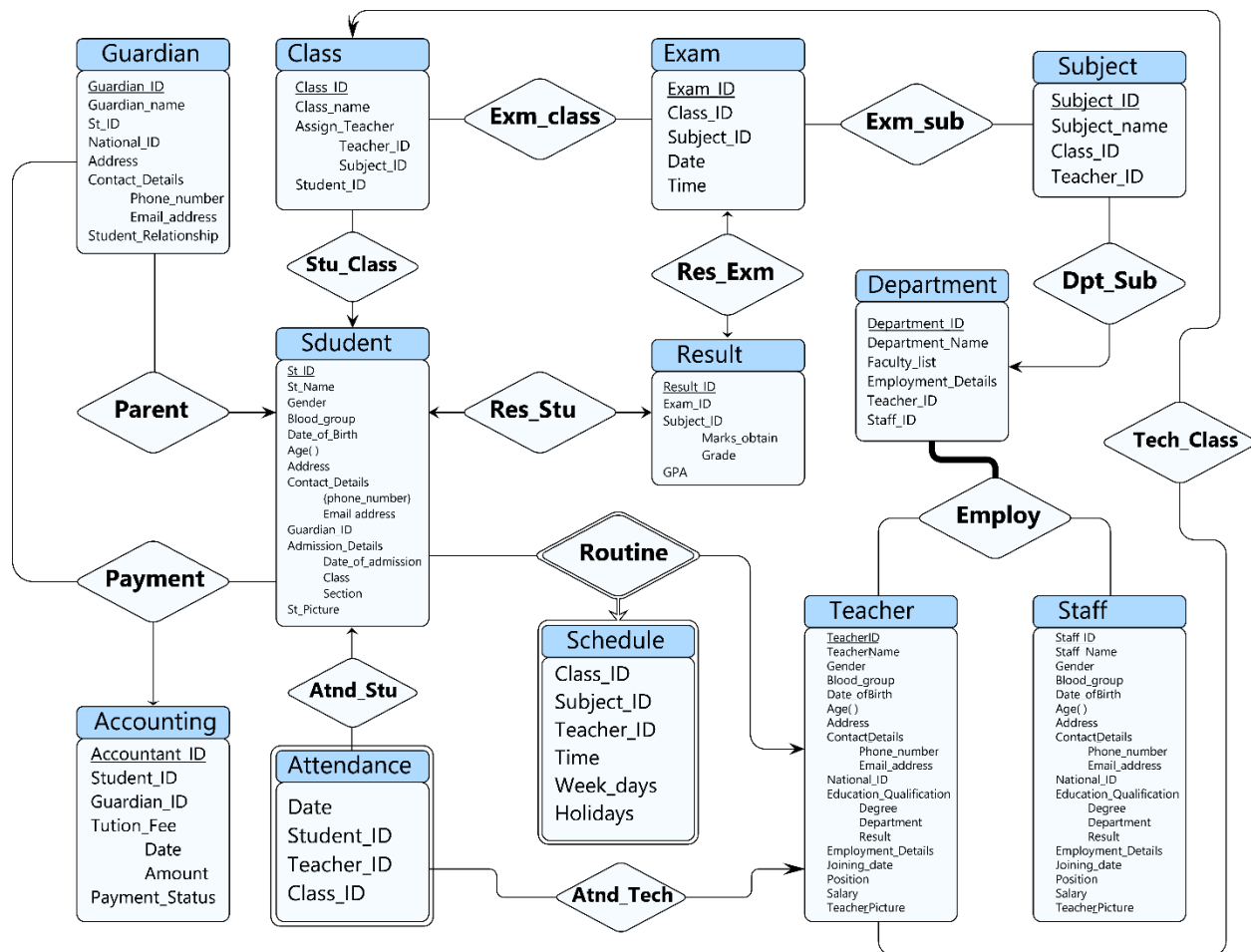
❖ **Accounting - Student (One-to-Many)**



Participation:

- Partial Participation: **department**
- Full Participation: **Student, Guardian, Teacher, Result, Staff, Department, Subject, Class, Exam, Attendance, Schedule, Accounting**

ER-Diagram:



Schema for this.

- **Department** (Department_ID, Department_Name, Faculty_list, Employment_Details, Teacher_ID, Staff_ID)
- **Teacher** (Teacher_ID, Teacher_Name, Gender, Blood_group, Date_of_Birth, Age(), Address, Phone_number, Email_address, National_ID, SSC, HSC, BSc, Department, Result, Joining_date, Position, Salary, Teacher_Picture)
- **Subject** (Subject_ID, Subject_name, Class_ID, Teacher_ID)
- **Class** (Class_ID, Class_name, Assign_Teacher, Teacher_ID, Subject_ID)
- **Exam** (Exam_ID, Class_ID, Subject_ID, Date, Time)
- **Result** (Result_ID, Exam_ID, Subject_ID, Marks_obtain, Grade, GPA)
- **Attendance** (Date, Student_ID, Teacher_ID, Class_ID)

- **Schedule** (Class_ID, Subject_ID, Teacher_ID, Time, Week_days, Holidays)
- **Accounting** (Accountant_ID, Student_ID, Guardian_ID, Tution_Fee, Date, Amount, Payment_Status)
- **Guardian** (Guardian_ID, Guardian_name, St_ID, National_ID, Address, Phone_number, Email_address, Student_Relationship)
- **Student** (St_ID, St_Name, Gender, Blood_group, Date_of_Birth, Age(), Address, Contact_Details, Guardian_ID, Admission_Details, St_Picture)